

The Basics of Volatility Trading

Volatility trading first attracted mathematically oriented traders who noticed that the market's prediction of forthcoming volatility—for example, implied volatility—was substantially out of line with what one might reasonably expect should happen. Moreover, many of these traders (market-makers, arbitrageurs, and others) had found great difficulties with keeping a “delta neutral” position neutral. Seeking a better way to trade without having a market opinion on the underlying security, they turned to volatility trading. This is not to suggest that volatility trading eliminates all market risk, turning it all into volatility risk, for example. But it does suggest that a certain segment of the option trading population can handle the risk of volatility with more deference and aplomb than they can handle price risk.

Simply stated, it seems like a much easier task to predict volatility than to predict prices. That is said notwithstanding the great bull market of the 1990s, in which every investor who strongly participated certainly feels that *he* understands how to predict prices. Remember not to confuse brains with a bull market. Consider the chart in Figure 36-1. This seems as if it might be a good stock to trade: Buy it near the lows and sell it near the highs, perhaps even selling it short near the highs and covering when it later declines. It appears to have been in a trading range for a long time, so that after each purchase or sale, it returns at least to the midpoint of its trading range and sometimes even continues on to the other side of the range. There is no scale on the chart, but that doesn't change the fact that it appears to be a tradeable entity. In fact, this is a chart of *implied volatility* of the options on a major U.S. corporation. It really doesn't matter which one (it's IBM), because the implied volatility chart of nearly every stock, index, or futures contract has a similar pattern—a trading range. The only time that implied volatility will totally break out of its

FIGURE 36-1.
A sample chart.



“normal” range is if something material happens to change the fundamentals of the way the stock moves—a takeover bid, for example, or perhaps a major acquisition or other dilution of the stock.

So, many traders observed this pattern and have become adherents of trying to predict volatility. Notice that if one is able to isolate volatility, he doesn’t care where the stock price goes—he is just concerned with buying volatility near the bottom of the range and selling it when it gets back to the middle or high end of the range, or vice versa. In real life, it is nearly impossible for a public customer to be able to isolate volatility so specifically. He will have to pay some attention to the stock price, but he still is able to establish positions in which the direction of the stock price is irrelevant to the outcome of the position. This quality is appealing to many investors, who have repeatedly found it difficult to predict stock prices. Moreover, an approach such as this should work in both bull and bear markets. Thus, volatility trading appeals to a great number of individuals. Just remember that, for you *personally* to operate a strategy properly, you must find that it appeals to your own philosophy of trading. Trying to use a strategy that you find uncomfortable will only lead to losses and frustration. So, if this somewhat neutral approach to option trading sounds interesting to you, then read on.

DEFINITIONS OF VOLATILITY

Volatility is merely the term that is used to describe how fast a stock, future, or index changes in price. When one speaks of volatility in connection with options, there are two

types of volatility that are important. The first is *historical volatility*, which is a measure of how fast the underlying instrument *has been* changing in price. The other is *implied volatility*, which is the option market's *prediction* of the volatility of the underlying over the life of the option. The computation and comparison of these two measures can aid immensely in predicting the forthcoming volatility of the underlying instrument—a crucial matter in determining today's option prices.

Historical volatility can be measured with a specific formula, as shown in the chapter on mathematical applications. It is merely the formula for standard deviation as contained in most elementary books on statistics. The important point to understand is that it is an exact calculation, and there is little debate over how to compute historical volatility. It is not important to know what the actual measurement means. That is, if one says that a certain stock has a historical volatility of 20%, that by itself is a relatively meaningless number to anyone but an ardent statistician. However, it *can* be used for comparative purposes.

The standard deviation is expressed as a percent. One can determine that the historical volatility of the broad stock market has usually been in the range of 15% to 20%. A very volatile stock might have an historical volatility in excess of 100%. These numbers can be compared to each other, so that one might say that a stock with the latter historical volatility is five times more volatile than the "stock market." So, the historical volatility of one instrument can be compared with that of another instrument in order to determine which one is more volatile. That in itself is a useful function of historical volatility, but its uses go much further than that.

Historical volatility can be measured over different time periods to give one a sense of how volatile the underlying has been over varying lengths of time. For example, it is common to compute a 10-day historical volatility, as well as a 20-day, 50-day, and even 100-day. In each case, the results are annualized so that one can compare the figures directly.

Consider the chart in Figure 36-2. It shows a stock (although it could be a futures contract or index, too) that was meandering in a rather tight range for quite some time. At the point marked "A" on the chart, it was probably at its least volatile. At that time, the 10-day volatility might have been something quite low, say 20%. The price movements directly preceding point A had been very small. However, prior to that time the stock had been more volatile, so longer-term measures of the historical volatility would have shown higher numbers. The possible measures of historical volatility, then *at point A*, might have been something like:

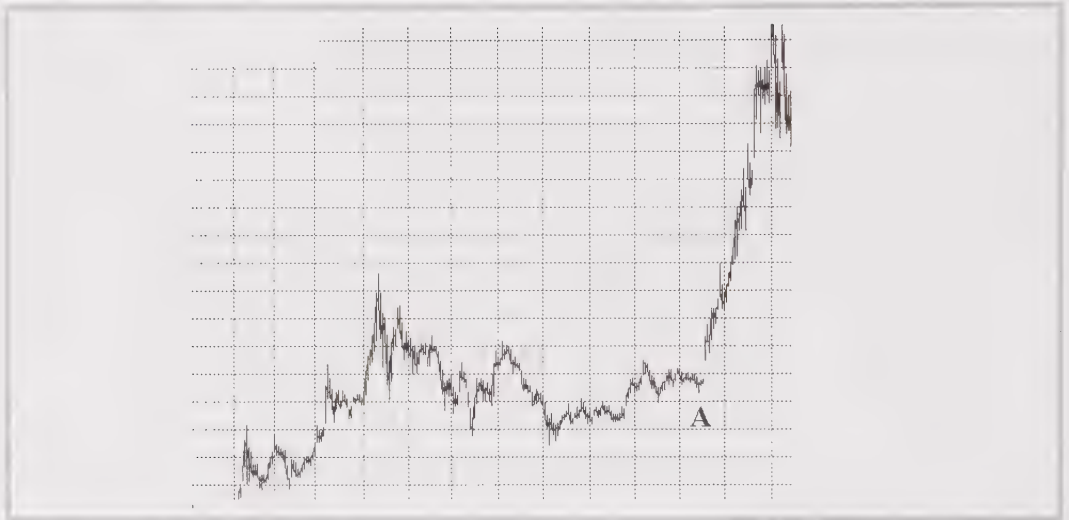
10-day historical volatility: 20%

20-day historical volatility: 23%

50-day historical volatility: 35%

100-day historical volatility: 45%

FIGURE 36-2.
Sample stock chart.



A pattern of historical volatilities of this sort describes a stock that has been slowing down lately.

Its price movements have been less extreme in the near term.

Again referring to Figure 36-2, note that shortly after point A, the stock jumped much higher over a short period of time. Price action like this increases the implied volatility dramatically. And, at the far right edge of the chart, the stock had stopped rising but was swinging back and forth in far more rapid fashion than it had been at most other points on the chart. Violent action in a back-and-forth manner can often produce a higher historical volatility reading than a straight-line move can; it's just the way the numbers work out. So, by the far right edge of the chart, the 10-day historical volatility would have increased rather dramatically, while the longer-term measures wouldn't be so high because they would still contain the price action that occurred prior to point A.

At the far right edge of Figure 36-2, these figures might apply:

10-day historical volatility: 80%

20-day historical volatility: 75%

50-day historical volatility: 60%

100-day historical volatility: 55%

With this alignment of historical volatilities, one can see that the stock has been more volatile recently than in the more distant past. In Chapter 38 on the distribution of

stock prices, we will discuss in some detail just which one, if any, of these historical volatilities one should use as “the” historical volatility input into option and probability models. We need to be able to make volatility estimates in order to determine whether or not a strategy might be successful, and to determine whether the current option price is a relatively cheap one or a relatively expensive one. For example, one can’t just say, “I think XYZ is going to rise at least 18 points by February expiration.” There needs to be some basis in fact for such a statement and, lacking inside information about what the company might announce between now and February, that basis should be statistics in the form of volatility projections.

Historical volatility is, of course, useful as an input to the (Black–Scholes) option model. In fact, the volatility input to any model is crucial because the volatility component is such a major factor in determining the price of an option. Furthermore, historical volatility is useful for more than just estimating option prices. It is necessary for making stock price projections and calculating distributions, too, as will be shown when those topics are discussed later. Any time one asks the question, “What is the probability of the stock moving from here to there, or of exceeding a particular target price?” the answer is heavily dependent on the volatility of the underlying stock (or index or futures).

It is obvious from the above example that historical volatility can change dramatically for any particular instrument. Even if one were to stick with just one measure of historical volatility (the 20-day historical is commonly the most popular measure), it changes with great frequency. Thus, one can never be certain that basing option price predictions or stock price distributions on the current historical volatility will yield the “correct” results. Statistical volatility may change as time goes forward, in which case your projections would be incorrect. Thus, it is important to make projections that are on the conservative side.

ANOTHER APPROACH: GARCH

GARCH stands for *Generalized Autoregressive Conditional Heteroskedasticity*, which is why it’s shortened to GARCH. It is a technique for forecasting volatility that some analysts say produces better projections than using historical volatility alone or implied volatility alone. GARCH was created in the 1980s by specialists in the field of econometrics. It incorporates both historical and implied volatility, plus one can throw in a constant (“fudge factor”). In essence, though, the user of GARCH volatility models has to make some predictions or decisions about the weighting of the factors used for the estimate. By its very nature, then, it can be just as vague as the situations described in the previous section.

The model can “learn,” though, if applied correctly. That is, if one makes a volatility prediction for today (using GARCH, let’s say), but it turns out that the actual volatility was

lower, then when you make the volatility prediction for tomorrow, you'll probably want to adjust it downward, using the experience of the real world, where you see volatility declining. This also incorporates the common-sense notion that volatility tends to remain the same; that is, tomorrow's volatility is likely to be much like today's. Of course, that's a little bit like saying tomorrow's weather is likely to be the same as today's (which it is, two-thirds of the time, according to statistics). It's just that when a tornado hits, you have to realize that your forecast could be wrong. The same thing applies to GARCH volatility projections. They can be wrong, too.

So, GARCH does not do a perfect job of estimating and forecasting volatility. In fact, it might not even be superior, from a strategist's viewpoint, to using the simple minimum/maximum techniques outlined in the previous section. It is really best geared to predicting short-term volatility and is favored most heavily by dealers in currency options who must adjust their markets constantly. For longer-term volatility projections, which is what a *position trader* of volatility is interested in, GARCH may not be all that useful. However, it is considered state-of-the-art as far as volatility predicting goes, so it has a following among theoretically oriented traders and analysts.

MOVING AVERAGES

Some traders try to use moving averages of daily composite implied volatility readings, or use a smoothing of recent past historical volatility readings to make volatility estimates. As mentioned in the chapter on mathematical applications, once the composite daily implied volatility has been computed, it was recommended that a smoothing effect be obtained by taking a moving average of the 20 or 30 days' implied volatilities. In fact, an *exponential* moving average was recommended, because it does not require one to keep accessing the last 20 or 30 days' worth of data in order to compute the moving average. Rather, the most recent exponential moving average is all that's needed in order to compute the next one.

IMPLIED VOLATILITY

Implied volatility has been mentioned many times already, but we want to expand on its concept before getting deeper into its measure and uses later in this section. Implied volatility pertains only to options, although one can aggregate the implied volatilities of the various options trading on a particular underlying instrument to produce a single number, which is often referred to as the composite implied volatility of the underlying.

At any one point in time, a trader knows for certain the following items that affect an option's price: stock price, strike price, time to expiration, interest rate, and dividends. The only remaining factor is volatility—in fact, *implied* volatility. It is the big “fudge factor” in option trading. If implied volatility is too high, options will be overpriced. That is, they will be relatively expensive. On the other hand, if implied volatility is too low, options will be cheap or underpriced. The terms “overpriced” and “underpriced” are not really used by theoretical option traders much anymore, because their usage implies that one knows what the option *should* be worth. In the modern vernacular, one would say that the options are trading with a “high implied volatility” or a “low implied volatility,” meaning that one has some sense of where implied volatility has been in the past, and the current measure is thus high or low in comparison.

Essentially, implied volatility is the option market's guess at the forthcoming statistical volatility of the underlying over the life of the option in question. If traders believe that the underlying will be volatile over the life of the option, then they will bid up the option, making it more highly priced. Conversely, if traders envision a nonvolatile period for the stock, they will not pay up for the option, preferring to bid lower; hence the option will be relatively low-priced. The important thing to note is that traders normally do *not* know the future. They have no way of knowing, for sure, how volatile the underlying is going to be during the life of the option.

Having said that, it would be unrealistic to assume that inside information does not leak into the marketplace. That is, if certain people possess nonpublic knowledge about a company's earnings, new product announcement, takeover bid, and so on, they will aggressively buy or bid for the options and that will increase implied volatility. So, in certain cases, when one sees that implied volatility has shot up quickly, it is perhaps a signal that some traders do indeed know the future—at least with respect to a specific corporate announcement that is about to be made.

However, most of the time there is not anyone trading with inside information. Yet, every option trader—market-maker and public alike—is forced to make a “guess” about volatility when he buys or sells an option. That is true because the price he pays is heavily influenced by his volatility estimate (whether or not he realizes that he is, in fact, making such a volatility estimate). As you might imagine, most traders have no idea what volatility is going to be during the life of the option. They just pay prices that seem to make sense, perhaps based on historical volatility. Consequently, today's implied volatility may bear no resemblance to the actual statistical volatility that later unfolds during the life of the option.

For those who desire a more mathematical definition of implied volatility, consider this:

Opt price = $f(\text{Stock price, Strike price, Time, Risk-free rate, Volatility, Dividends})$

Furthermore, suppose that one knows the following information:

XYZ price: 52
April 50 call price: 6
Time remaining to April expiration: 36 days
Dividends: \$0.00
Risk-free interest rate: 5%

This information, which is available for every option at any time, simply from an option quote, gives us everything except the implied volatility. So what volatility would one have to plug in the Black–Scholes model (or whatever model one is using) to make the model give the answer 6 (the current price of the option)? That is, what volatility is necessary to solve the equation?

$$6 = f(52, 50, 36 \text{ days}, 5\%, \text{Volatility}, \$0.00)$$

Whatever volatility is necessary to make the model yield the current market price (6) as its value, is the implied volatility for the XYZ April 50 call. In this case, if you're interested, the implied volatility is 75.4%. The actual process of determining implied volatility is an iterative one. There is no formula, *per se*. Rather, one keeps trying various volatility estimates in the model until the answer is close enough to the market value.

THE VOLATILITY OF VOLATILITY

In order to discuss the implied volatility of a particular entity—stock, index, or futures contract—one generally refers to the implied volatility of individual options or perhaps the composite implied volatility of the entire option series. This is generally good enough for strategic comparisons. However, it turns out that there might be other ways to consider looking at implied volatility. In particular, one might want to consider how wide the *range* of implied volatility is—that is, how volatile the individual implied volatility numbers are.

It is often conventional to talk about the *percentile of implied volatility*. That is a way to rank the current implied volatility reading with past readings for the same underlying instrument.

However, a fairly important ingredient is missing when percentiles are involved. One can't really tell if "cheap" options are cheap as a practical matter. That's because one doesn't know how tightly packed together the past implied volatility readings are. For example, if one were to discover that the entire past range of implied volatility for XYZ stretched only from 39% to 45%, then a current reading of 40%, while low, might not

seem all that attractive. That is, if the first percentile of XYZ options were at an implied volatility reading of 39% and the 100th percentile were at 45%, then a reading of 40% is really quite mundane. There just wouldn't be much room for implied volatility to increase on an absolute basis. Even if it rose to the 100th percentile, an individual XYZ option wouldn't gain much value, because its implied volatility would only be increasing from about 40% to 45%.

However, if the distribution of past implied volatility is *wide*, then one can truly say the options are cheap if they are currently in a low percentile. Suppose, rather than the tight range described above, that the range of past implied volatilities for XYZ instead stretched from 35% to 90%—that the first percentile for XYZ implied volatility was at 35% and the 100th percentile was at 90%. Now, if the current reading is 40%, there is a large range above the current reading into which the options could trade, thereby potentially increasing the value of the options if implied volatility moved up to the higher percentiles.

What this means, as a practical matter, is that one not only needs to know the current percentile of implied volatility, but he also needs to know the *range* of numbers over which that percentile was derived. If the range is wide, then an extreme percentile truly represents a cheap or expensive option. But if the range is tight, then one should probably not be overly concerned with the current percentile of implied volatility.

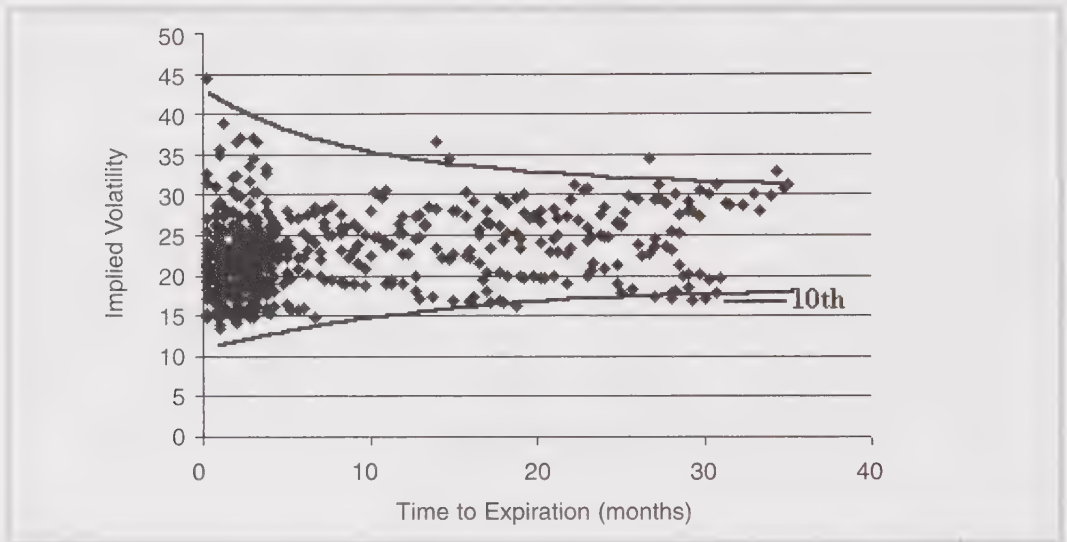
Another facet of implied volatility that is often overlooked is how it ranges with respect to the time left in the option. This is particularly important for traders of LEAPS (long-term) options, for the range of implied volatility of a LEAPS option will not be as great as that of a short-term option. In order to demonstrate this, the implied volatilities of OEX options, both regular and LEAPS, were charted over several years. The resulting scatter diagram is shown in Figure 36-3.

Two curved lines are drawn on Figure 36-3. They contain most of the data points. One can see from these lines that the range of implied volatility for near-term options is greater than it is for longer-term options. For example, the implied volatility readings on the far left of the scatter diagram range from about 14% to nearly 40% (ignore the one outlying point). However, for longer-term options of 24 months or more, the range is about 17% to 32%. While OEX options have their own idiosyncracies, this scatter diagram is fairly typical of what we would see for any stock or index option.

One conclusion that we can draw from this is that LEAPS option implied volatilities just don't change nearly as much as those of short-term options. That can be an important piece of information for a LEAPS option trader especially if he is comparing the LEAPS implied volatility with a *composite* implied volatility or with the *historical* volatility of the underlying.

Once again, consider Figure 36-3. While it is difficult to discern from the graph alone, the 10th percentile of OEX composite implied volatility, using all of the data points given, is 17%. The line that marks this level (the tenth percentile) is noted on the right

FIGURE 36-3.
Implied volatilities of OEX options over several years.



side of the scatter diagram. It is quite easy to see that the LEAPS options rarely trade at that low volatility level.

In Figure 36-3, the distance between the curved lines is much greater on the left side (i.e., for shorter-term options) than it is on the right side (for longer-term options). Thus, it's difficult for the longer-term options to register either an extremely high or extremely low implied volatility reading, when *all* of the options are considered. Consequently, LEAPS options will rarely appear “cheap” when one looks at their percentile of implied volatility, including all the short-term options, too.

One might say that, if he were going to buy long-term options, he should look only at the size of the volatility range on the right side of the scatter diagram. Then, he could make his decision about whether the options are cheap or not by only comparing the current reading to past readings of long-term options. This line of thinking, though, is somewhat fallacious reasoning, for a couple of reasons: First, if one holds the option for any long period of time, the volatility range will widen out and there is a chance that implied volatility could drop substantially. Second, the long-term volatility range might be so small that, even though the options are initially cheap, quick increase in implied volatility over several deciles might not translate into much of a gain in price in the short term.

It's important for anyone using implied volatility in his trading decisions to understand that the range of past implied volatilities is important, and to realize that the volatility range expands as time shrinks.

IS IMPLIED VOLATILITY A GOOD PREDICTOR OF ACTUAL VOLATILITY?

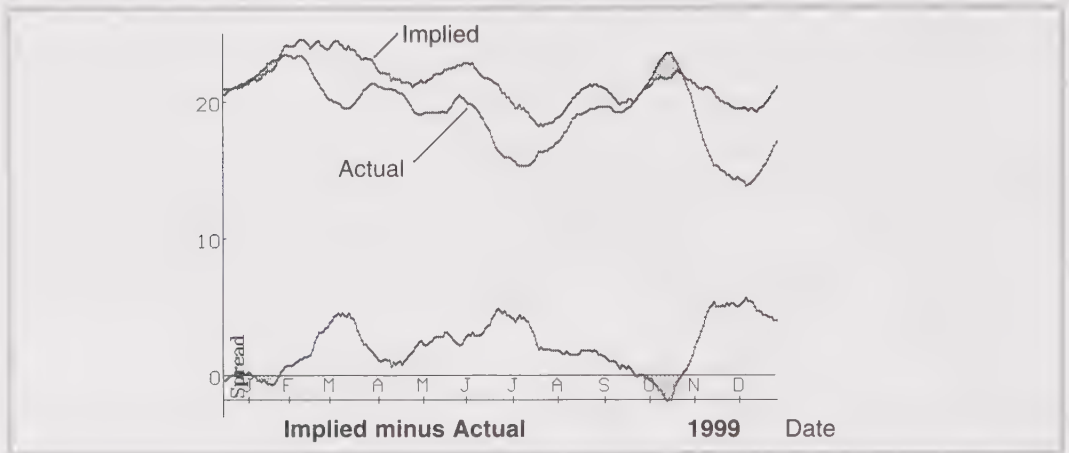
The fact that one can calculate implied volatility does not mean that the calculation is a good estimate of forthcoming volatility. As stated above, the marketplace does not really know how volatile an instrument is going to be, any more than it knows the forthcoming price of the stock. There are clues, of course, and some general ways of estimating forthcoming volatility, but the fact remains that sometimes options trade with an implied volatility that is quite a bit out of line with past levels. Therefore, implied volatility may be considered to be an inaccurate estimate of what is really going to happen to the stock during the life of the option. Just remember that implied volatility is a forward-looking estimate, and since it is based on traders' suppositions, it can be wrong—just as any estimate of future events can be in error.

The question posed above is one that should probably be asked more often than it is: “Is implied volatility a good predictor of actual volatility?” Somehow, it seems logical to assume that implied and historical (actual) volatility will converge. That’s not really true, at least not in the short term. Moreover, even if they *do* converge, which one was right to begin with—implied or historical? That is, did implied volatility move to get more in line with actual movements of the underlying, or did the stock’s movement speed up or slow down to get in line with implied volatility?

To illustrate this concept, a few charts will be used that show the comparison between implied and historical volatility. Figure 36-4 shows information for the OEX Index. In general, OEX options are overpriced. See the discussion in Chapter 29. That is, implied volatility of OEX options is almost always higher than what actual volatility turns out to be. Consider Figure 36-4. There are three lines in the figure: (a) implied volatility, (b) actual volatility, and (c) the difference between the two. There is an important distinction here, though, as to what comprises these curves:

- (a) The implied volatility curve depicts the 20-day moving average of daily composite implied volatility readings for OEX. That is, each day one number is computed as a composite implied volatility for OEX for that day. These implied volatility figures are computed using the averaging formula shown in the chapter on mathematical applications, whereby each option’s implied volatility is weighted by trading volume and by distance in- or out-of-the-money, to arrive at a single composite implied volatility reading for the trading day. To smooth out those daily readings, a 20-day simple moving average is used. This daily implied volatility of OEX options encompasses *all* the OEX options, so it is different from the Volatility Index (VIX), which uses only the options closest to the money. By using all of the options, a slightly different volatility figure is arrived at, as compared to VIX, but a chart of the two would show

FIGURE 36-4.
OEX implied versus historical volatility.



similar patterns. That is, peaks in implied volatility computed using all of the OEX options occur at the same points in time as peaks in VIX.

- (b) The *actual* volatility on the graph is a little different from what one normally thinks of a historical volatility. It is the 20-day historical volatility, computed 20 days *later* than the date of the implied volatility calculation. Hence, points on the implied volatility curve are matched with a 20-day historical volatility calculation *that was made 20 days later*. Thus, the two curves more or less show the prediction of volatility and what actually happened over the 20-day period. These actual volatility readings are smoothed as well, with a 20-day moving average.
- (c) The difference between the two is quite simple, and is shown as the bottom curve on the graph. A “zero” line is drawn through the difference.

When this “difference line” passes through the zero line, the projection of volatility and what actually occurred 20 days later were equal. If the difference line is above the zero line, then implied volatility was too high; the options were overpriced. Conversely, if the difference line is below the zero line, then actual volatility turned out to be greater than implied volatility had anticipated. The options were underpriced in that case. Those latter areas are shaded in Figure 36-4. Simplistically, you would want to own options during the shaded periods on the chart, and would want to be a seller of options during the non-shaded areas.

Note that Figure 36-4 indeed confirms the fact that OEX options are consistently overpriced. Very few charts are as one-dimensional as the OEX chart, where the options

were so consistently overpriced. Most stocks find the difference line oscillating back and forth about the zero mark. Consider Figures 36-5 and 36-6. Figure 36-5 shows a chart similar to Figure 36-4, comparing actual and implied volatility, and their difference, for a particular stock. Figure 36-6 shows the price graph of that same stock, overlaid on implied volatility, during the period up to and including the heavy shading.

The volatility comparison chart (Figure 36-5) shows several shaded areas, during which the stock was more volatile than the options had predicted. Owners of options profited during these times, provided they had a more or less neutral outlook on the stock. Figure 36-6 shows the stock's performance up to and including the March–April 1999 period—the largest shaded area on the chart. Note that implied volatility was quite low before the stock made the strong move from 10 to 30 in little more than a month. These graphs are taken from actual data and demonstrate just how badly out of line implied volatility can be. In February and early March 1999, implied volatility was at or near the lowest levels on these charts. Yet, by the end of March, a major price explosion had begun in the stock, one that tripled its value in just over a month. Clearly, implied volatility was a poor predictor of forthcoming actual volatility in this case.

What about later in the year? In Figure 36-5, one can observe that implied and actual volatility oscillated back and forth quite a few times during the rest of 1999. It might appear that these oscillations are small and that implied volatility was actually

FIGURE 36-5.
Implied versus historical volatility of a stock.

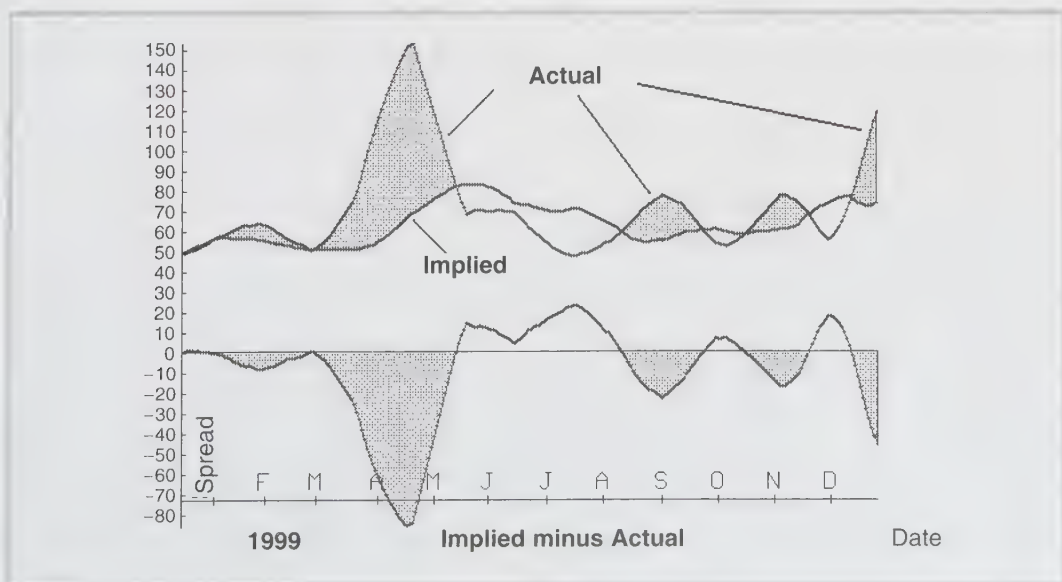
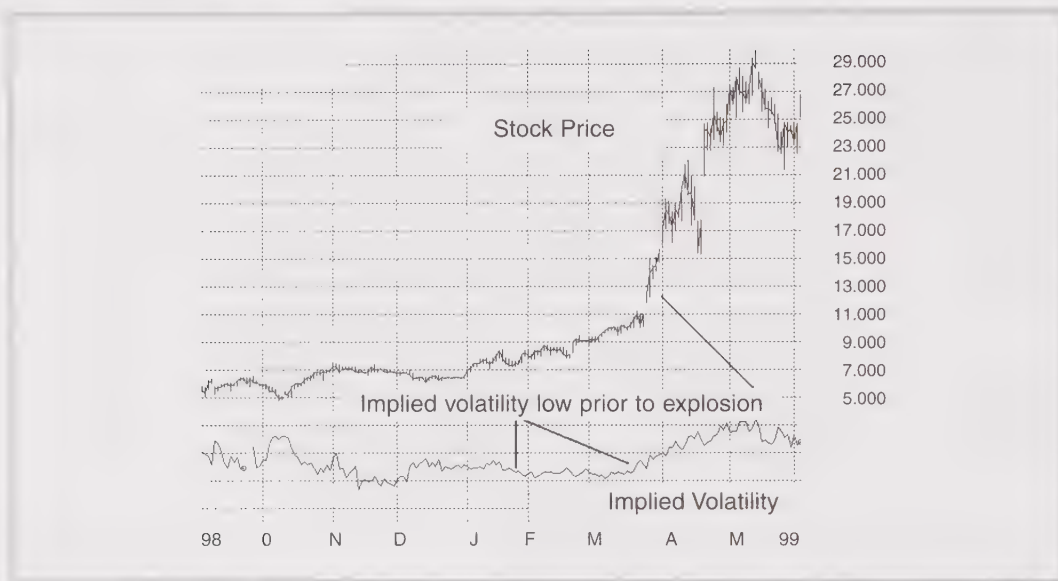


FIGURE 36-6.
The price graph of the stock.



doing a pretty good job of predicting actual volatility, at least until the final spike in December 1999. However, looking at the scale on the left-hand side of Figure 36-5, one can see that implied volatility was trying to remain in the 50% to 60% range, but actual volatility kept bolting higher rather frequently.

One more example will be presented. Figures 36-7 and 36-8 depict another stock and its volatilities. On the left half of each graph, implied volatility was quite high. It was higher than actual volatility turned out to be, so the difference line in Figure 36-7 remains above the zero line for several months. Then, for some reason, the option market decided to make an adjustment, and implied volatility began to drop. Its lowest daily point is marked with a circle in Figure 36-8, and the same point in time is marked with a similar circle in Figure 36-7. At that time, options traders were “saying” that they expected the stock to be very tame over the ensuing weeks. Instead, the stock made two quick moves, one from 15 down to 11, and then another back up to 17. That movement jerked actual volatility higher, but implied volatility remained rather low. After a period of trading between 13 and 15, during which time implied volatility remained low, the stock finally exploded to the upside, as evidenced by the spikes on the right-hand side of both Figures 36-7 and 36-8. Thus, implied volatility was a poor predictor of actual volatility for most of the time on these graphs. Moreover, implied volatility remained low at the right-hand side of the charts (January 2000) even though the stock doubled in the course of a month.

FIGURE 36-7.
Implied versus historical volatility of a stock.

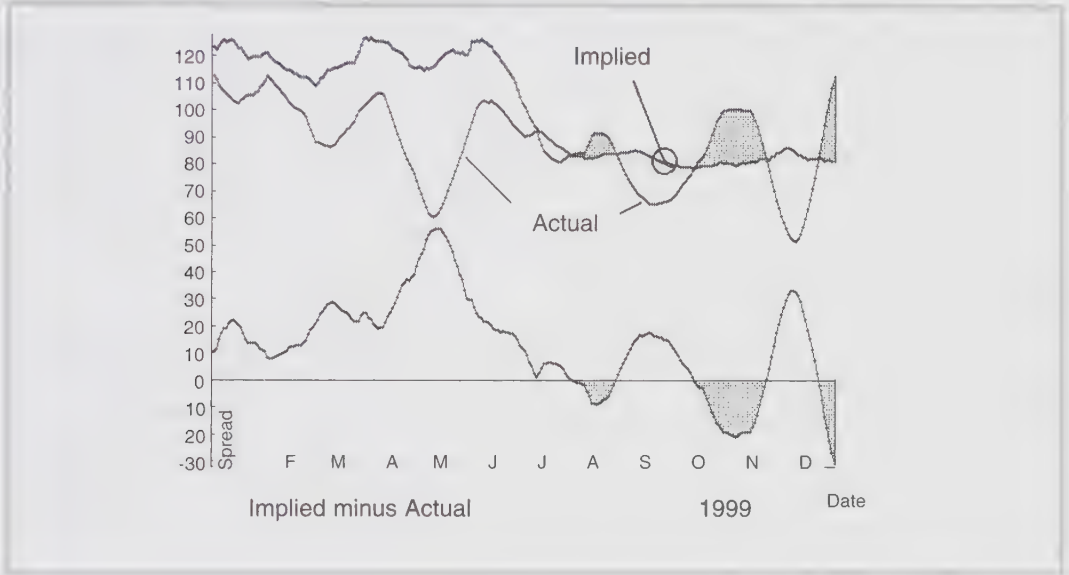
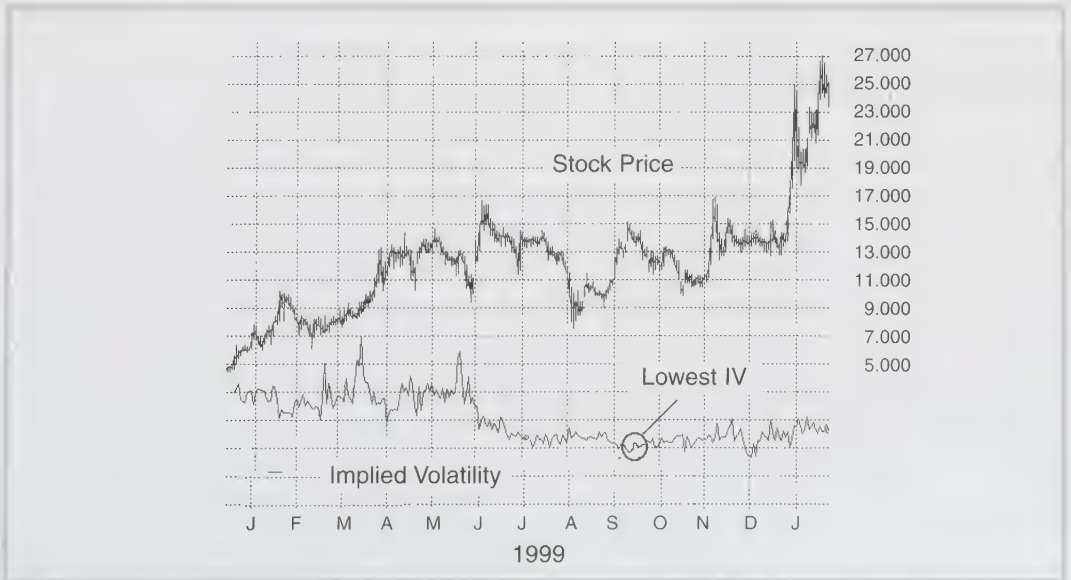


FIGURE 36-8.
The price graph of the stock.



The important thing to note from these figures is that they clearly show that implied volatility is really not a very good predictor of the actual volatility that is to follow. If it were, the difference line would hover near zero most of the time. Instead, it swings back and forth wildly, with implied volatility over- or underestimating actual volatility by quite wide levels. Thus, the current estimates of volatility by traders (i.e., implied volatility) can actually be quite wrong.

Conversely, one could also say that historical volatility is not a great predictor of volatility that is to follow, either, especially in the short term. No one really makes any claims that it is a good predictor, for historical volatility is merely a reflection of what has happened in the past. All we can say for sure is that implied and historical volatility tend to trade within a range.

One thing that does stand out on these charts is that implied volatility seems to fluctuate *less* than actual volatility. That seems to be a natural function of the volatility predictive process. For example, when the market collapses, implied volatilities of options rise only modestly. In other words, option traders and market-makers are predicting volatility when they price options, and one tends to make a prediction that is somewhat “middle of the road,” since an extreme prediction is more likely to be wrong. Of course, it turns out to be wrong anyway, since actual volatility jumps around quite rapidly.

The few charts that have been presented here don’t constitute a rigorous study upon which to draw the conclusion that implied volatility is a poor predictor of actual volatility, but it is this author’s firm opinion that that statement is true. A graduate student looking for a master’s thesis topic could take it from here.

VOLATILITY TRADING

As a result of the fact that implied volatility can sometimes be at irrational extremes, options may sometimes trade with implied volatilities that are significantly out of line with what one would normally expect. For example, suppose a stock is in a relatively non-volatile period, like the price of the stock in Figure 36-2, just before point A on the graph. During that time, option sellers would probably become more aggressive while option buyers, who probably have been seeing their previous purchases decaying with time, become more timid. As a result, option prices drop. Alternatively stated, implied volatility drops. When implied volatilities are decreasing, option sellers are generally happy (and may often become more aggressive), while option buyers are losing money (and may often tend to become more timid). This is just a function of looking at the profit and loss statements in one’s option account. But anyone who took a longer backward look at the volatility of the stock in Figure 36-2 would see that it had been much more volatile in the

past. Consequently, he might decide that the implied volatility of the options had gotten too low and he would be a buyer of options.

It is the volatility trader's objective to spot situations when implied volatility is possibly or probably erroneous and to take a position that would profit when the error is brought to light. Thus, the volatility trader's main objective is spotting situations when implied volatility is overvalued or undervalued, irrespective of his outlook for the underlying stock itself. In some ways, this is not so different from the fundamental stock analyst who is attempting to spot overvalued or undervalued stocks, based on earnings and other fundamentals.

From another viewpoint, volatility trading is also a contrarian theory of investing. That is, when everyone else thinks the underlying is going to be nonvolatile, the volatility trader buys volatility. When everyone else is selling options and option buyers are hard to find, the volatility trader steps up to buy options. Of course, some rigorous analysis must be done before the volatility trader can establish new positions, but when those situations come to light, it is most likely that he is taking positions opposite to what "the masses" are doing. He will be buying volatility when the majority has been selling it (or at least, when the majority is refusing to buy it), and he will be selling volatility when everyone else is panicking to buy options, making them quite expensive.

WHY DOES VOLATILITY REACH EXTREMES?

One can't just buy every option that he considers to be cheap. There must be some consideration given to what the probabilities of stock movement are. Even more important, one can't just sell every option that he values as expensive. There may be valid reasons why options become expensive, not the least of which is that someone may have inside information about some forthcoming corporate news (a takeover or an earnings surprise, for example).

Since options offer a good deal of leverage, they are an attractive vehicle to anyone who wants to make a quick trade, especially if that person believes he knows something that the general public doesn't know. Thus, if there is a leak of a takeover rumor—whether it be from corporate officers, investment bankers, printers, or accountants—whoever possesses that information may quite likely buy options aggressively, or at least *bid* for them. Whenever demand for an option outstrips supply—in this case, the major supplier is probably the market-maker—the options quickly get more expensive. That is, implied volatility increases.

In fact, there are financial analysts and reporters who look for large increases in trading volume as a clue to which stocks might be ready to make a big move. Invariably, if the trading volume has increased and if implied volatility has increased as well, it is a

good warning sign that someone with inside information is buying the options. In such a case, it might *not* be a good idea to sell volatility, even though the options are mathematically expensive.

Sometimes, even more minor news items are known in advance by a small segment of the investing community. If those items will be enough to move the stock even a couple of points, those who possess the information may try to buy options in advance of the news. Such minor news items might include the resignation or firing of a high-ranking corporate officer, or perhaps some strategic alliance with another company, or even a new product announcement.

The seller of volatility can watch for two things as warning signs that perhaps the options are “predicting” a corporate event (and hence should be avoided as a “volatility sale”). Those two things are a dramatic increase in option volume or a sudden jump in implied volatility of the options. One or both can be caused by traders with inside information trying to obtain a leveraged instrument in advance of the actual corporate news item being made public.

A SUDDEN INCREASE IN OPTION VOLUME OR IMPLIED VOLATILITY

The symptoms of insider trading, as evidenced by a large increase in option trading activity, can be recognized. Typically, the majority of the increased volume occurs in the near-term option series, particularly the at-the-money strike and perhaps the next strike out-of-the-money. The activity doesn't cease there, however. It propagates out to other option series as market-makers (who by the nature of their job function are *short* the near-term options that those with insider knowledge are buying) snap up everything on the books that they can find. In addition, the market-makers may try to entice others, perhaps institutions, to sell some expensive calls against a portion of their institutional stock holdings. Activity of this sort should be a warning sign to the volatility seller to stand aside in this situation.

Of course, on any given day there are many stocks whose options are extraordinarily active, but the increase in activity doesn't have anything to do with insider trading. This might include a large covered call write or maybe a large put purchase established by an institution as a hedge against an existing stock position, or a relatively large conversion or reversal arbitrage established by an arbitrageur, or even a large spread transaction initiated by a hedge fund. In any of these cases, option volume would jump dramatically, but it wouldn't mean that anyone had inside knowledge about a forthcoming corporate event. Rather, the increases in option trading volume as described in this paragraph are merely functions of the normal workings of the marketplace.

What distinguishes these arbitrage and hedging activities from the machinations of insider trading is: (1) There is little propagation of option volume into other series in the

“benign” case, and (2) the stock price itself may languish. However, when true insider activity is present, the market-makers react to the aggressive nature of the call buying. These market-makers know they need to hedge themselves, because they do not want to be short naked call options in case a takeover bid or some other news spurs the stock dramatically higher. As mentioned earlier, they try to buy up any other options offered in “the book,” but there may not be many of those. So, as a last result, the way they reduce their negative position delta is to *buy stock*. Thus, if the options are active and expensive, and if the stock is rising too, you probably have a reasonably good indication that “someone knows something.” However, if the options are expensive but none of the other factors are present, especially if the stock is *declining* in price—then one might feel more comfortable with a strategy of selling volatility in this case.

However, there is a case in which options might be the object of pursuit by someone with insider knowledge, yet not be accompanied by heavy trading volume. This situation could occur with illiquid options. In this case, a floor broker holding the order of those with insider information might come into the pit to buy options, but the market-makers may not sell them many, preferring to raise their offering price rather than sell a large quantity. If this happens a few times in a row, the options will have gotten very expensive as the floor broker raises his bid price repeatedly, but only buys a few contracts each time. Meanwhile, the market-maker keeps raising his offering price.

Eventually, the floor broker concludes that the options are too expensive to bother with and walks away. Perhaps his client then buys stock. In any case, what has happened is that the options have gotten very expensive as the bids and offers were repeatedly raised, but not much option volume was actually traded because of the illiquidity of the contracts. Hence the normal warning light associated with a sudden increase in option volume would not be present. In this case, though, a volatility seller should still be careful, because he does not want to step in to sell calls right before some major corporate news item is released. The clue here is that implied volatility *exploded* in a short period of time (one day, or actually less time), and that alone should be enough warning to a volatility seller.

The point that should be taken here is that when options suddenly become very expensive, especially if accompanied by strong stock price movement and strong stock volume, there may very well be a good reason why that is happening. That reason will probably become public knowledge shortly in the form of a news event. In fact, a major market-maker once said he believed that *most* increases in implied volatility were eventually justified—that is, some corporate news item was released that made the stock jump. Hence, a volatility seller should avoid situations such as these. Any sudden increase in implied volatility should probably be viewed as a potential news story in the making. These situations are not what a neutral volatility seller wants to get into.

On the other hand, if options have become expensive *as a result of* corporate news,

then the volatility seller can feel more comfortable making a trade. Perhaps the company has announced poor earnings and the stock has taken a beating while implied volatility rose. In this situation, one can assess the information and analyze it clearly; he is not dealing with some hidden facts known to only a few insider traders. With clear analysis, one might be able to develop a volatility selling strategy that is prudent and potentially profitable.

Another situation in which options become expensive in the wake of market action is during a bear market in the underlying. This can be true for indices, stocks, and futures contracts. The Crash of '87 is an extreme example, but implied volatility shot through the roof during the crash. Other similar sharp market collapses—such as October 1989, October 1997, and August–September 1998—caused implied volatility to jump dramatically. In these situations, the volatility seller knows why implied volatility is high. Given that fact, he can then construct positions around a neutral strategy or around his view of the future. The time when the volatility seller must be careful is when the options are expensive and no one seems to know why. That's when insider trading may be present, and that's when the volatility seller should defer from selling options.

CHEAP OPTIONS

When options are cheap, there are usually far less discernible reasons why they have become cheap. An obvious one may be that the corporate structure of the company has changed; perhaps it is being taken over, or perhaps the company has acquired another company nearly its size. In either case, it is possible that the combined entity's stock will be less volatile than the original company's stock was. As the takeover is in the process of being consummated, the implied volatility of the company's options will drop, giving the false impression that they are cheap.

In a similar vein, a company may mature, perhaps issuing more shares of stock, or perhaps building such a good earnings stream that the stock is considered less volatile than it formerly was. Some of the Internet companies will be classic cases: In the beginning they were high-flying stocks with plenty of price movement, so the options traded with a relatively high degree of implied volatility. However, as the company matures, it buys other Internet companies and then perhaps even merges with a large, established company (America Online and Time-Warner Communications, for example). In these cases, actual (statistical) volatility will diminish as the company matures, and implied volatility will do the same. On the surface, a buyer of volatility may see the reduced volatility as an attractive buying situation, but upon further inspection he may find that it is justified. If the decrease in implied volatility seems justified, a buyer of volatility should ignore it and look for other opportunities.

All volatility traders should be suspicious when volatility seems to be extreme—either too expensive or too cheap. The trader should investigate the possibilities as to *why* volatility is trading at such extreme levels. In some cases, the supply and demand of the public just pushes the options to extreme levels; there is nothing more involved than that. Those are the best volatility trading situations. However, if there is a hint that the volatility has gotten to an extreme reading because of some logical (but perhaps nonpublic) reason, then the volatility trader should be suspicious and should probably avoid the trade. Typically this happens with expensive options.

Buyers of volatility really have little to fear if they miscalculate and thus buy an option that appears inexpensive but turns out not to be, in reality. The volatility buyer might lose money if he does this, and overpaying for options constantly will lead to ruin, but an occasional mistake will probably not be fatal.

Sellers of volatility, however, have to be a lot more careful. One mistake could be the last one. Selling naked calls that seem terrifically expensive by historic standards could be ruinous if a takeover bid subsequently emerges at a large premium to the stock's current price. Even put sellers must be careful, although a lot of traders think that selling naked puts is safe because it's the same as buying stock. But who ever said buying stock wasn't risky? If the stock collapses—falling from 80, say, to 15 or 20, as Oxford Health did, or from 30 to 2 as Sunrise Technology did—then a put seller will be buried. Since the risk of loss from naked option selling is large, one could be wiped out by a huge gap opening. That's why it's imperative to study *why* the options are expensive before one sells them. If it's known, for example, that a small biotech company is awaiting FDA trial results in two weeks, and all the options suddenly become expensive, the volatility seller should *not* attempt to be a hero. It's obvious that at least some traders believe that there is a chance for the stock to gap in price dramatically. It would be better to find some other situation in which to sell options.

The seller of futures options or index options should be cautious too, although there can't be takeovers in those markets, nor can there be a huge earnings surprise or other corporate event that causes a big gap. The futures markets, though, do have things like crop reports and government economic data to deal with, and those can create volatile situations, too. The bottom line is that volatility selling—even *hedged* volatility selling—can be taxing and aggravating if one has sold volatility in front of what turns out to be a news item that justifies the expensive volatility.

SUMMARY

Volatility trading is a predictable way to approach the market, because volatility almost invariably trades in a range and therefore its value can be estimated with a great deal

more precision than can the actual prices of the underlyings. Even so, one must be careful in his approach to volatility trading, because diligent research is needed to determine if, in fact, volatility is “cheap” or “expensive.” As with any systematic approach to the market, if one is sloppy about his research, he cannot expect to achieve superior results. In the next few chapters, a good deal of time will be spent to give the reader a good understanding of how volatility affects positions and how it can be used to construct trades with positive expected rates of return.